Capital Budgeting

Chapter 8

Introduction to Financial Management (Pope)

Internal Investment

Recall from Chapter 1, financial decisions are either investment or financing decisions

- Investment decisions in corporate finance are often called ***capital budgeting decisions***
- Capital budgeting = long-term decisions for a firm
- How do you make the most appropriate capital budgeting decision?

Net Present Value

$$NPV = \sum_{t=0}^{T} \frac{C_t}{(1+r)^t}$$

$$NPV = \frac{C_0}{(1+r)^0} + \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \cdots \frac{C_t}{(1+r)^t}$$

- *r* is the opportunity cost of capital (i.e. cost of financing)
- C_t is the period-specific cash flow in time period *t*
- *What is this?*

Net Present Value

$$NPV = \sum_{i=0}^{T} \frac{C_i}{(1+r)^t}$$

$$NPV = \frac{C_0}{(1+r)^0} + \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \cdots \frac{C_t}{(1+r)^t}$$

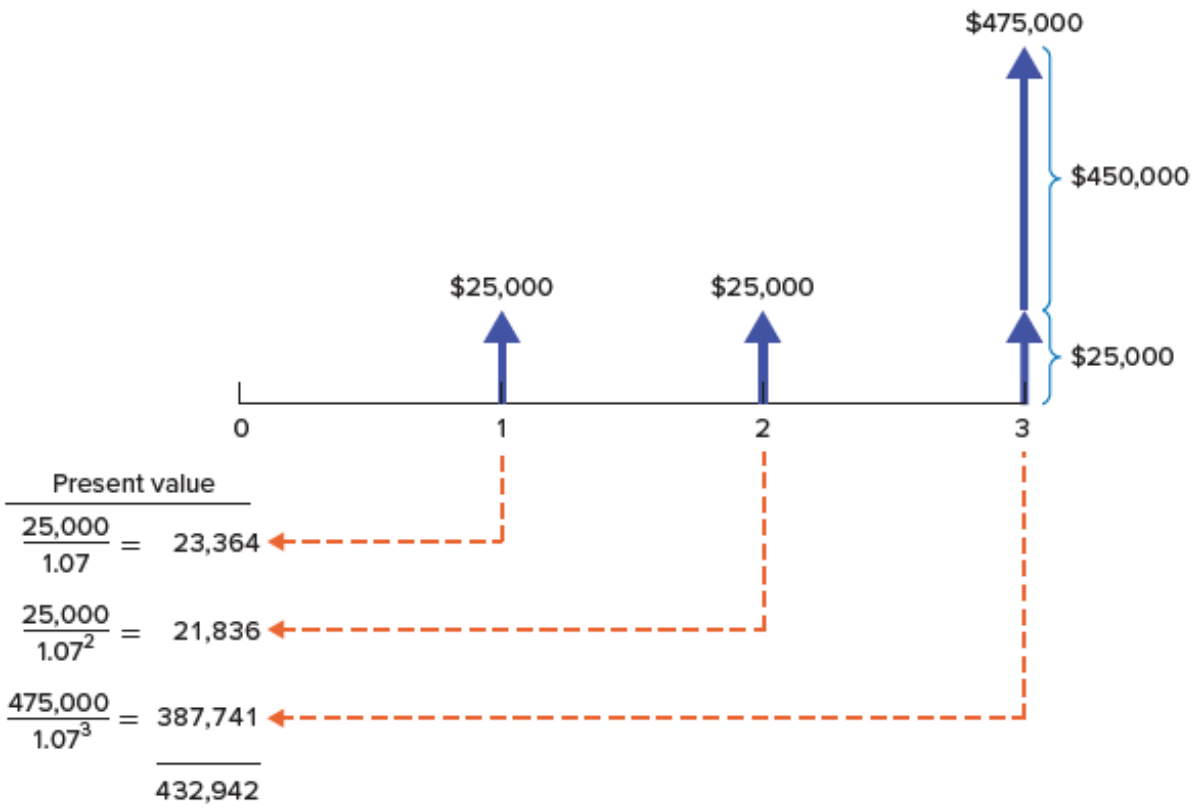
- *If NPV > 0, accept the project; if NPV < 0, reject the project*
- Why?

Example 1:

Your firm is considering an investment into an office building. There are tenants that will generate $25,000 per year in cash flows for 3 years. At the end of 3 years, you think your firm can sell the building for $450,000. If your firm's opportunity cost of capital is 7% and plan to invest $375,000 in the building, is this a worthwhile project?

Example 1:

Your firm is considering an investment into an office building. There are tenants that will generate $25,000 per year in cash flows for 3 years. At the end of 3 years, you think your firm can sell the building for $450,000. If your firm's opportunity cost of capital is 7% and plan to invest $375,000 in the building, is this a worthwhile project?



Example 1:

Your firm is considering an investment into an office building. There are tenants that will generate $25,000 per year in cash flows for 3 years. At the end of 3 years, you think your firm can sell the building for $450,000. If your firm's opportunity cost of capital is 7% and plan to invest $375,000 in the building, is this a worthwhile project?

$$NPV = C_0 + \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \cdots \frac{C_t}{(1+r)^t}$$

$$NPV = -375{,}000 + \frac{25{,}000}{(1+0.07)^1} + \frac{25{,}000}{(1+0.07)^2} + \frac{475{,}000}{(1+0.07)^3}$$

$$NPV = 57{,}942$$

Internal Rate of Return

$$C_0 = \frac{C_1}{(1+IRR)^1} + \frac{C_2}{(1+IRR)^2} + \cdots \frac{C_t}{(1+IRR)^t}$$

The rate of return implied by a project's cost and future cash flows

- *IRR* must be calculated
- IRR can also be called the discounted cash-flow (DCF) rate of return
- IRR rule: ***If IRR > opportunity cost of capital, accept the project; if IRR < opportunity cost of capital, reject the project***

IRR and NPV

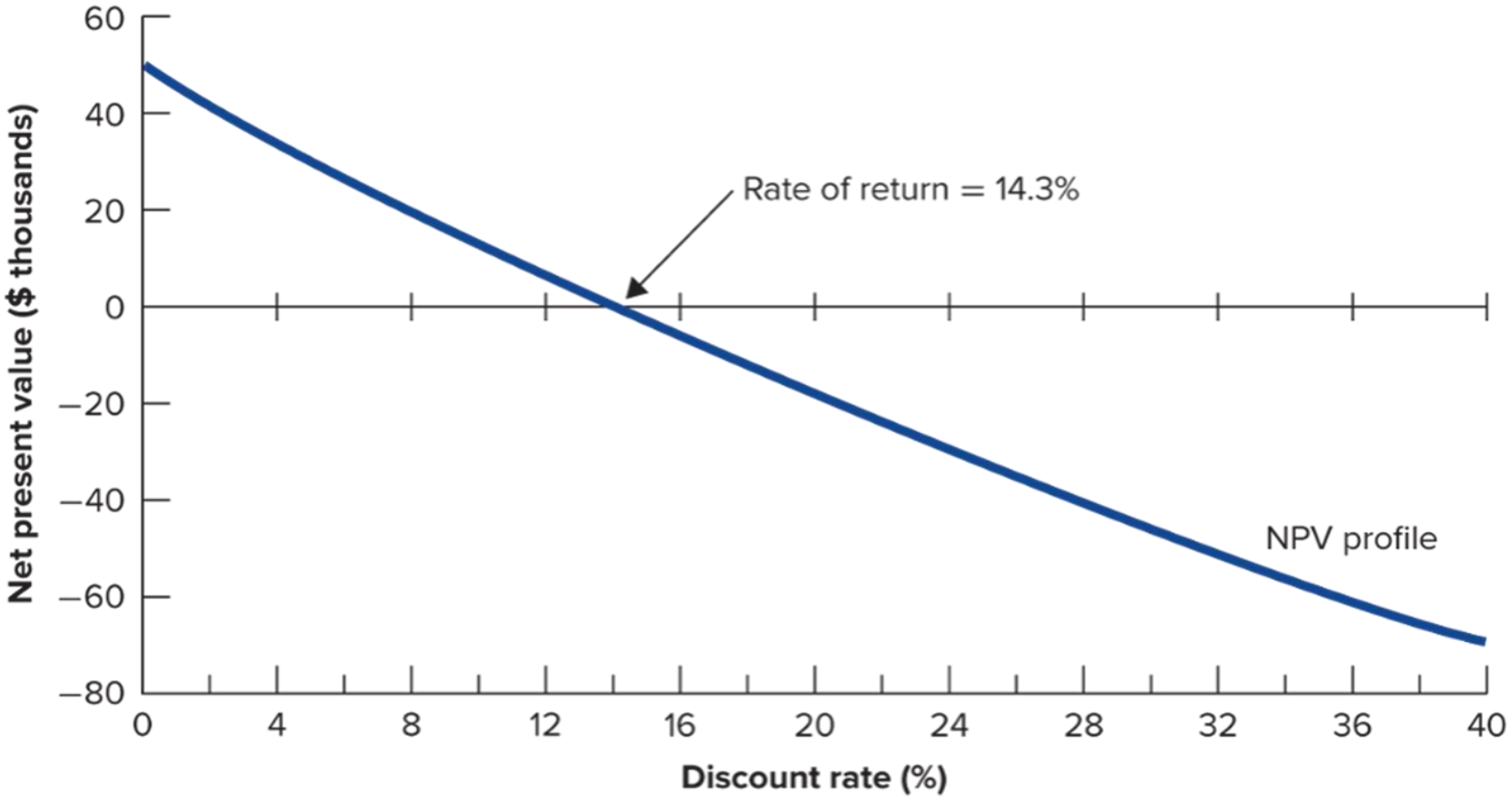
$$NPV = C_0 + \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \cdots \frac{C_t}{(1+r)^t} = 0$$

$$C_0 = \frac{C_1}{(1+IRR)^1} + \frac{C_2}{(1+IRR)^2} + \cdots \frac{C_t}{(1+IRR)^t}$$

- *IRR* is the discount rate that sets NPV equal to zero
- ***If IRR > opportunity cost of capital, generally NPV is positive; if IRR < opportunity cost of capital, generally NPV is negative***

NPV Profile

An NPV profile shows the NPV of an investment with different discount rates. Here, as the discount rate is increased, the NPV falls.



Example 2:

Recall the office building investment from before. There are tenants that will generate $25,000 per year in cash flows for 3 years. At the end of 3 years, you think your firm can sell the building for $450,000. If your firm's opportunity cost of capital is 7% and plan to invest $375,000 in the building, what is the projects internal rate of return (IRR)?

Example 2:

Recall the office building investment from before. There are tenants that will generate $25,000 per year in cash flows for 3 years. At the end of 3 years, you think your firm can sell the building for $450,000. If your firm's opportunity cost of capital is 7% and plan to invest $375,000 in the building, what is the projects internal rate of return (IRR)?

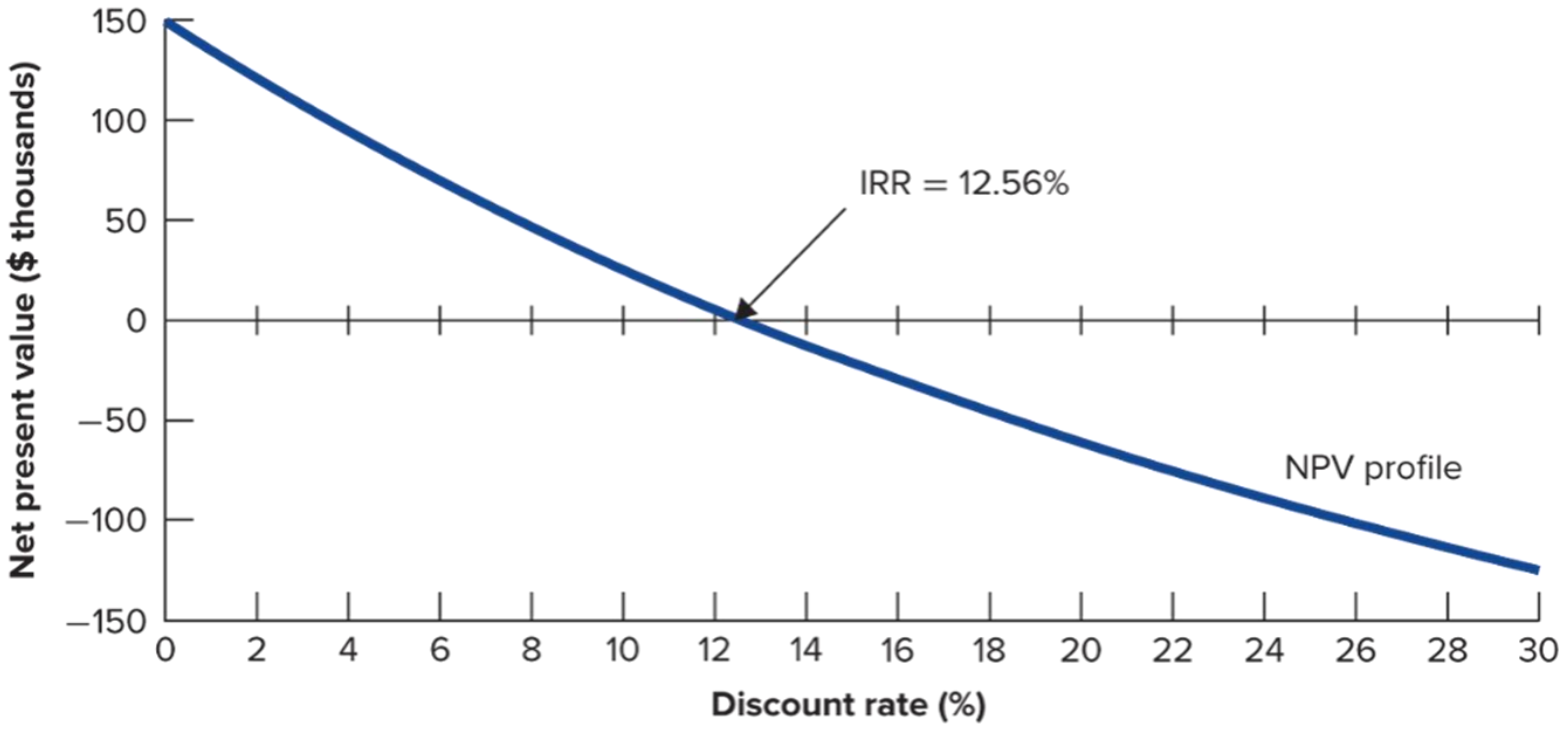
$$C_0 = \frac{C_1}{(1+IRR)^1} + \frac{C_2}{(1+IRR)^2} + \cdots \frac{C_t}{(1+IRR)^t}$$

$$375{,}000 = \frac{25{,}000}{(1+IRR)^1} + \frac{25{,}000}{(1+IRR)^2} + \frac{475{,}000}{(1+IRR)^3}$$

$$IRR = 0.1256 = 12.56\%$$

Example 2:

Recall the office building investment from before. There are tenants that will generate $25,000 per year in cash flows for 3 years. At the end of 3 years, you think your firm can sell the building for $450,000. If your firm's opportunity cost of capital is 7% and plan to invest $375,000 in the building, what is the projects internal rate of return (IRR)?



Special Cases

Case 1: Mutually exclusive projects

- With NPV, choosing project with highest NPV is often the best choice (but not always)
- With IRR, choosing the highest IRR can lead to poor decision making
- Pay attention to either the **timing of cash flows (projects have different lives)** or **large differences in outlays**

Example 3: Example 3: Determine the Cross-over Rate

Recall the office building investment from before ($25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years, opportunity cost of capital is 7% and plan to invest $375,000). Your company's initial proposal (your predecessor) was to invests only $350,000, receives no rent payment and sells the property in 1 year for $400,000. What is the initial project's internal rate of return (IRR) and which should your firm choose?

Example 3: Example 3: Determine the Cross-over Rate

Recall the office building investment from before ($25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years, opportunity cost of capital is 7% and plan to invest $375,000). Your company's initial proposal (your predecessor) was to invests only $350,000, receives no rent payment and sells the property in 1 year for $400,000. What is the initial project's internal rate of return (IRR) and which should your firm choose?

$$C_0 = \frac{C_1}{(1+IRR)^1} + \frac{C_2}{(1+IRR)^2} + \cdots \frac{C_t}{(1+IRR)^t}$$

$$350{,}000 = \frac{400{,}000}{(1+IRR_I)^1}$$

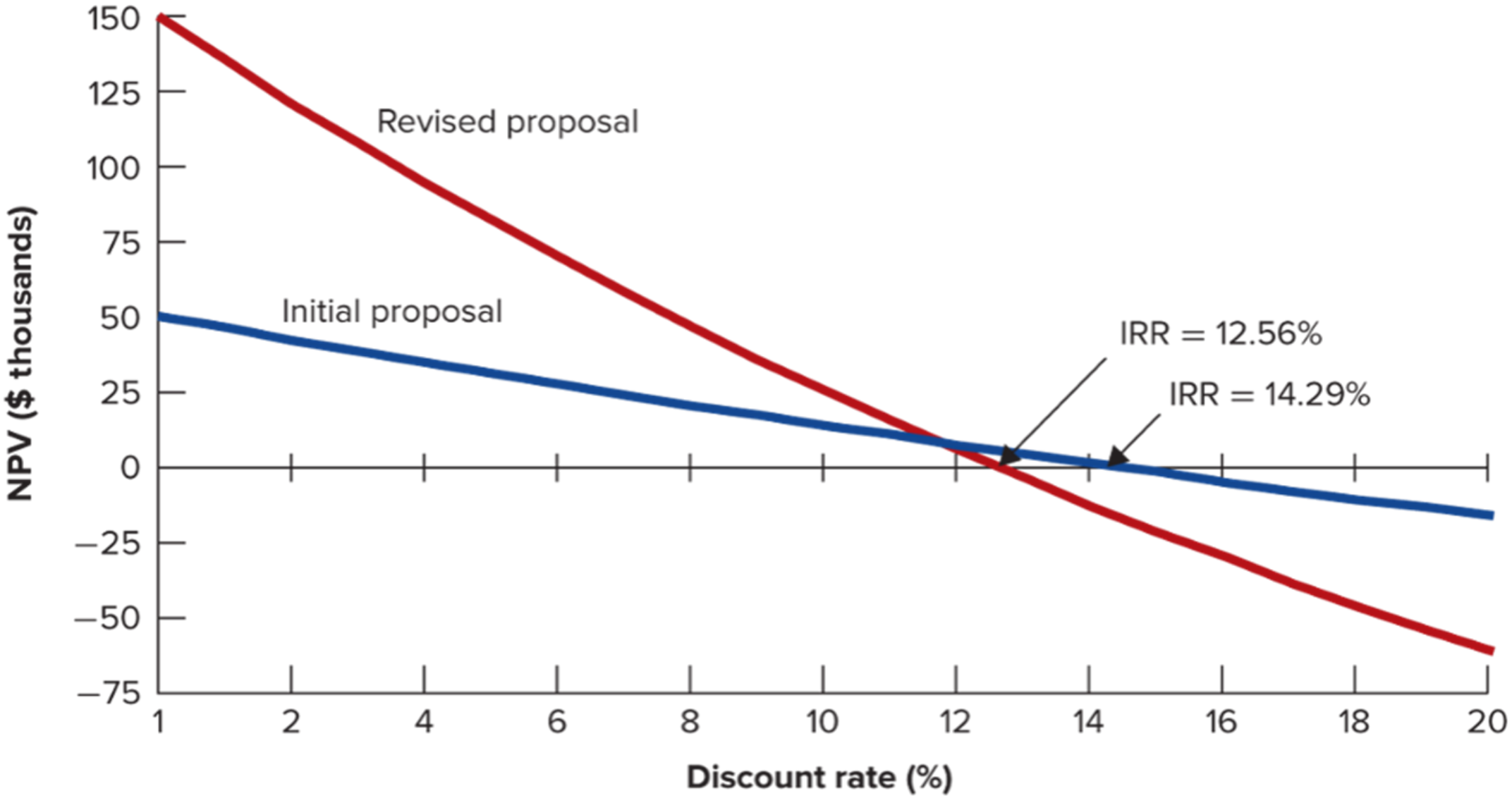
$$IRR_I = 0.1429 = 14.29\%$$

$$IRR_R = 12.56\%$$

Example 3: Example 3: Determine the Cross-over Rate

Recall the office building investment from before ($25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years, opportunity cost of capital is 7% and plan to invest $375,000). Your company's initial proposal (your predecessor) was to invests only $350,000, receives no rent payment and sells the property in 1 year for $400,000. What is the initial project's internal rate of return (IRR) and which should your firm choose?

Project	C_0	C_1	C_2	C_3	IRR	NPV @ 7%
Revised proposal	–375	25	25	475	12.56%	$57,942
Initial proposal	–350	400			14.29%	$23,832


150
125
100
75
50
25
0
−25
−50
−75
NPV ($ thousands)
Revised proposal
Initial proposal
IRR = 12.56%
IRR = 14.29%
1
2
4
6
8
10
12
14
16
18
20
Discount rate (%)

Example 3: Determine the Cross-over Rate

Recall the office building investment from before ($25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years, opportunity cost of capital is 7% and plan to invest $375,000). Your company's initial proposal (your predecessor) was to invests only $350,000, receives no rent payment and sells the property in 1 year for $400,000. What is the initial project's internal rate of return (IRR) and which should your firm choose?

Project	C_0	C_1	C_2	C_3	IRR	NPV @ 7%
Revised proposal	–375	25	25	475	12.56%	$57,942
Initial proposal	–350	400			14.29%	$23,832
Revised - Initial	-25	-375	25	475	11.72%	

Example 4: Equivalent Annual Annuity/Cost

Samuel's Doughnut Shop is considering the purchase of one of two machines. Machine A is a dough mixing machine which cost $45,000 and brings an annual cost savings of $15,000 for 5 years. Machine B is an icing machine which cost $20,000 and brings an annual cost savings of $12,000 for 3 years. If your cost of capital is 8%, which of the two projects should you accept?

Example 4: Equivalent Annual Annuity/Cost

Samuel's Doughnut Shop is considering the purchase of one of two machines. Machine A is a dough mixing machine which cost $45,000 and brings an annual cost savings of $15,000 for 5 years. Machine B is an icing machine which cost $20,000 and brings an annual cost savings of $12,000 for 3 years. If your cost of capital is 8%, which of the two projects should you accept?

$$NPV_A = \$14{,}890.65$$

$$NPV_B = \$10{,}925.16$$

Even though Machine A has the higher NPV, it includes additional cash flows that machine B does not and hence we need a new way to value the projects. Take each NPV and turn it into an equivalent annuity payment:

$$EAA = \frac{NPV}{PVAF} = \frac{NPV}{\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right)}$$

Example 4: Equivalent Annual Annuity/Cost

Samuel's Doughnut Shop is considering the purchase of one of two machines. Machine A is a dough mixing machine which cost $45,000 and brings an annual cost savings of $15,000 for 5 years. Machine B is an icing machine which cost $20,000 and brings an annual cost savings of $12,000 for 3 years. If your cost of capital is 8%, which of the two projects should you accept?

$$NPV_A = \$14{,}890.65$$

$$NPV_B = \$10{,}925.16$$

$$EAA_A = \frac{14{,}890.65}{\left(\frac{1}{0.08} - \frac{1}{0.08(1+0.08)^5}\right)} = 3{,}729.46$$

$$EAA_B = \frac{10{,}925.16}{\left(\frac{1}{0.08} - \frac{1}{0.08(1+0.08)^3}\right)} = 4{,}239.33$$

Special Cases

Case 2: Lending or Borrowing

- Rule for IRR: NPV must be falling
- Not the case when lending and borrowing (positive initial CF, negative future CFs)

Case 3: Multiple IRRs

- Occurs with non-normal CFs (change in sign of CF during stream)

Modified IRR

Modified IRR takes the cashflows of a project and "reinvests" them at the firm's cost of capital

- Avoids unrealistic return assumptions of traditional IRR
- Assumes firm will earn its cost of capital

$$MIRR = \left(\frac{FV_{inflows}}{PV_{outflows}}\right)^{1/N} - 1$$

Example 5

Recall the two office building investment from before 1) $25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years; 2) invest only $350,000 and sells the property in 1 year for $400,000. What are the modified internal rate of return (MIRR) of each project if the firm's cost of capital is 7%??

Example 5

Recall the two office building investment from before 1) $25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years; 2) invest only $350,000 and sells the property in 1 year for $400,000. What are the modified internal rate of return (MIRR) of each project if the firm's cost of capital is 7%?

Project	C_0	C_1	C_2	C_3	IRR	NPV @ 7%	MIRR
Revised proposal	–375	25	25	475	12.56%	$57,942	12.24%
Initial proposal	–350	400			14.29%	$23,832	14.29%

Profitability Index

$$PI = \frac{NPV}{initial\ cost}$$

NPV per dollar invested in a project

- Sometimes called the *Benefit-cost ratio*
- With a limited amount of capital and multiple projects, PI can be used as a ranking/decision criteria

Payback Period

Time until cash flows recover the initial investment of a project

- Accept a project if its payback period is less than pre-determined, specified cutoff period
- Ideal for small, short-term decision making
- Does not consider any projects after the payback period nor does it use TVM principles

Example 6:

Recall the two office building investment from before 1) $25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years; 2) invest only $350,000 and sells the property in 1 year for $400,000. What are the payback periods for both projects and which do you accept if your required payback cutoff is 2 years?

Example 6:

Recall the two office building investment from before 1) $25,000 per year in cash flows for 3 years, sell the building for $450,000 at the end of 3 years; 2) invest only $350,000 and sells the property in 1 year for $400,000. What are the payback periods for both projects and which do you accept if your required payback cutoff is 2 years?

Project	C_0	C_1	C_2	C_3	IRR	NPV @ 7%	Payback Period
Revised proposal	–375	25	25	475	12.56%	$57,942	2.68
Initial proposal	–350	400			14.29%	$23,832	0.875

Discounted Payback Period

Time until ***PRESENT VALUE*** of cash flows recover the initial investment of a project

- Each cash flow from the project is discounted to present value
- Incorporates TVM principals but still does not consider any projects after the payback period